Exp No:1.a

ANALYZE THE TREND OF DATA SCIENCE JOB POSTINGS OVER THE LAST DECADE

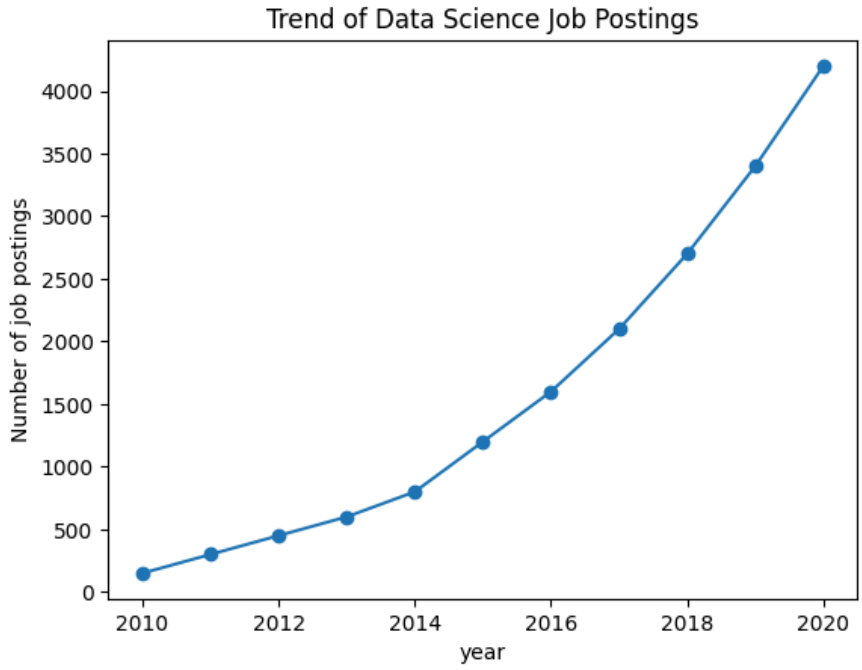
AIM:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
data={"year":list(range(2010,2021)),
      "Job postings":[150,300,450,600,800,1200,1600,2100,2700,3400,4200]}
df=pd.DataFrame(data)
plt.plot(df['year'],df['Job postings'],marker='o')
plt.title("Trend of Data Science Job Postings")
plt.xlabel('year')
plt.ylabel("Number of job postings")
plt.show()
```

OUTPUT:



RESULT:

The experiment has been carried out successfully.

Exp No:1.b

ANALYSE AND VISUALIZE THE DISTRIBUTION OF VARIOUS DATA SCIENCE ROLES(DATA ANALYST, DATA ENGINEER, DATA SCIENTIST, ETC.) FROM A DATASET

AIM:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

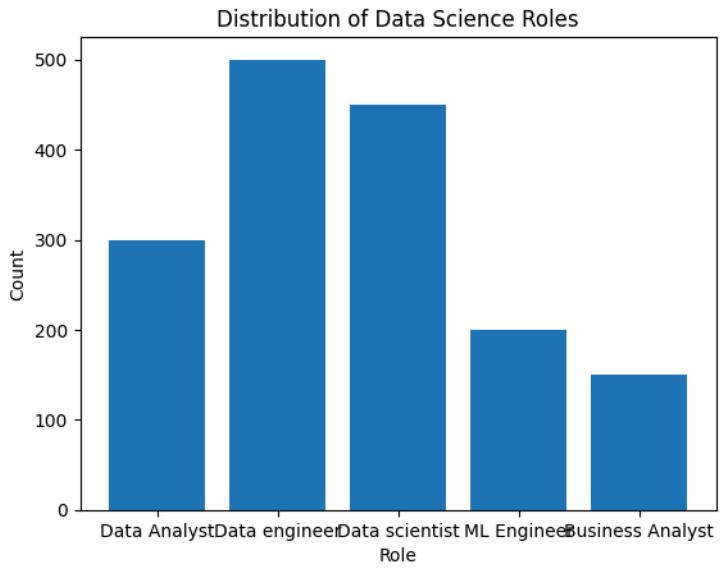
PROGRAM:

```
import matplotlib.pyplot as plt
roles=['Data Analyst', 'Data engineer','Data scientist', 'ML
Engineer','Business Analyst']
count=[300,500,450,200,150]
plt.bar(roles,count)
plt.title("Distribution of Data Science Roles")
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

SAMPLE INPUT :

```
roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business
Analyst.
counts = 300, 500, 450, 200, 150.
```

OUTPUT:



RESULT:

The experiment has been carried out successfully.

Exp No:1.c

CONDUCT AN EXPERIMENT TO DIFFERENTITATE STRUCTURED, UN-STRUCTURED AND SEMI STRUCTURED DATA BASED ON DATA SETS GIVEN

AIM:

Create small datasets for each type and explain their characteristics.

PROGRAM:

```
import pandas as pd
# Structured data example
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)
# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a
piece of text, an image,or a video file."
print("\nUnstructured Data:\n", unstructured_data)
# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height':
165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:
This is an example of unstructured data. It can be a piece of text, an image,or a video file.

Semi-structured Data:
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

RESULT:

The experiment has been carried out successfully.

Exp No:1.d

CONDUCT AN EXPERIMENT TO ENCRYPT AND DECRYPT GIVEN SENSITIVE DATA

AIM:

Use the cryptography library to encrypt and decrypt a piece of data.

PROGRAM:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data:
b'gAAAAABqs-aoO0KmmwNbQlMuwpJVhY1lcLr7XJOacn6FVa-c44biG8dczqIk23KBWTpeuptH
FkAJgdc3hbV76U0pkYbRWqpi5xBZZqogXl33x2zD_QJ01qaoUoeXHKNEwvdQqiYJzjhB'
```

RESULT:

The experiment has been carried out successfully.